

PlasticPaths - Project Requirements

1. Project Overview

PlasticPaths is a kid-friendly educational app that shows how floating litter moves through the ocean.

Users place different litter items on a coastal map, run a simulation, and explore their movement using a timeline. They can add collection points and compare how much litter gets intercepted.

An AI ocean guide explains the results through simple text and optional voice narration.

2. Target Audience

- Children approximately 8–12 years old.
- Teachers using interactive environmental lessons.
- Parents exploring ocean conservation with their children.

3. Project Goals

- Teach how ocean currents transport floating litter.
- Help users understand why cleanup location and timing matter.
- Encourage learning through predictions and experiments.
- Use real environmental data with clearly explained simulation assumptions.
- Provide a colourful, accessible, and engaging experience.

4. Main User Flow

1. Select a litter item, such as a bottle, bag, or foam packaging.
2. Click a water location to place it.
3. Predict where it might travel.
4. Run the simulation.
5. Use the timeline to see its movement over time.
6. Place a collection point and repeat the experiment.
7. Compare the results.
8. Read or listen to the ocean guide's explanation.

5. Functional Requirements

Feature	Requirement
Interactive map	Display one coastal region with water-direction arrows.
Litter placement	Allow users to select, place, and remove simulated litter.
Simulation	Calculate movement using ocean-current data.
Timeline	Support play, pause, restart, and time scrubbing.
Movement trails	Show the path each item has travelled.
Coastline handling	Mark litter as beached when it reaches land.
Collection points	Let users place collection points that intercept nearby litter.
Results	Show counts of floating, captured, beached, and outside-region items.
Comparison	Compare the same experiment with and without collection points.
AI explanations	Explain results using calculated metrics and retrieved educational facts.
Voice narration	Read explanations aloud when the user selects “Listen.”
Data attribution	Display environmental sources, dates, and simulation limitations.

6. UI and Design Requirements

- Bright ocean colours and illustrated litter items.
- Rounded cards and large buttons.
- Short labels and simple explanations.
- A friendly ocean mascot or guide.
- A large map and easy-to-use timeline.
- Positive feedback that encourages experimentation.
- Icons with text labels.
- Keyboard support and visible focus indicators.
- Optional audio with matching text.
- Reduced-motion support.
- A parent/teacher panel containing detailed assumptions and sources.

7. Technology Stack

Component	Technology	Purpose
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Frontend	React, TypeScript, Vite	Build the interactive application.
Styling	Tailwind CSS, shadcn/ui	Create accessible, playful controls.
Map	MapLibre GL JS, react-map-gl	Display the coastal map and placement tools.
Animation	Canvas, Motion	Animate litter, trails, and interface transitions.
Backend	Python, FastAPI	Run simulations and provide API endpoints.
Data processing	NumPy, SciPy, xarray	Read environmental data and calculate movement.
Geometry	Shapely	Detect coastlines and collection intersections.
Time-series database	Tiger Data / TimescaleDB	Store metadata, current samples, and trajectories.
AI retrieval	TiDB vector search	Retrieve relevant educational facts and similar scenarios.
AI explanations	Gemini API	Explain simulation results in child-friendly language.
Narration	ElevenLabs API	Generate optional spoken explanations.
Environmental data	UBC SalishSeaCast	Provide hourly modelled ocean-current data.

8. How the Technologies Work Together

Python and FastAPI: Receive the experiment settings and calculate litter trajectories.

Tiger Data: Store litter types and collection sites alongside timestamped currents and particle positions. This powers playback and result comparisons.

TiDB: Store a curated educational knowledge collection and use vector search to retrieve relevant information.

Gemini: Combine retrieved information with simulation results to explain what happened. Gemini does not calculate the litter's movement.

ElevenLabs: Convert the explanation into optional voice narration.

React: Display the map, animation, timeline, and results.

9. Simulation Approach

The simulation reads the water's speed and direction at each particle's location and moves it forward in small time steps.

The initial version will:

- Use one coastal region and a cached 24-hour data window.
- Model floating surface litter using simplified current-driven movement.
- Detect land crossings and collection points.
- Record positions for timeline playback.
- Use identical environmental inputs when comparing cleanup placements.

The UBC dataset contains near-surface depth-averaged currents. Wind, waves, sinking, and decomposition will not be modelled in the MVP.

Different litter illustrations will use the same drift behaviour unless documented item-specific parameters are added.

The app will clearly label results as a **simplified simulation**, rather than a validated real-world prediction.

10. Weekend MVP Scope

Required

- One coastal map.
- Three litter types.
- Placement and removal controls.
- Current-driven simulation.
- Playback and timeline.
- Coastline handling.
- Collection-point placement.
- Before-and-after comparison.
- TiDB educational retrieval.
- Gemini explanations.
- ElevenLabs narration.
- Kid-friendly styling.

Stretch Features

- Multiple languages.
- Prediction quizzes.
- Protected-habitat overlays.

- Wind effects.
- Additional regions.
- Shared classroom experiments.

11. Sustainability Goal Alignment

Goal 14: Life Below Water

Teach users how marine litter travels and let them explore ways to intercept it.

Goal 4: Quality Education

Provide accessible, interactive environmental learning.

Goal 17: Partnerships for the Goals

A possible extension would allow educators and cleanup organizations to contribute learning materials or share scenarios. This should only be claimed if a collaboration feature is implemented.

12. Target Hackathon Tracks

- Enactus UNSDG.
- Python Track.
- Best Use of Tiger Data.
- TiDB x AI Open Build.
- Best Use of Gemini API.
- Best Use of ElevenLabs.
- Best Design.
- Best Game, if challenge objectives and scoring are implemented.

Track eligibility and rules for entering multiple categories still need organizer confirmation.

Person	Owns	Main tasks	 
1. Map and animation	Interactive ocean experience	MapLibre map, litter placement, animated particles/trails, current arrows, collection-point placement.	
2. UI and educational experience	Everything around the map	Kid-friendly layout, litter selector, timeline controls, mascot panel, results/comparison cards, audio controls, styling.	
3. Simulation and Tiger Data	Numerical backend	Download/cache currents, Python simulation, coastline detection, interception logic, FastAPI simulation endpoints, Tiger Data storage.	
4. AI and narration	Educational backend	Curate sourced facts, TiDB vector search, Gemini explanations, ElevenLabs audio, explanation/narration endpoints.	